Pytorch

Deep backward schemes for high-dimensional nonlinear PDEs

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- Mathematical framework
- Formulation of the problem
- Application 1 : Study of the empirical error
- Application 2 : CVA estimation

BSDE definition

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{Q})$ be a filtered probability space.

A backward stochastic differential equation on $[0,T]$ is of the form :

$$\begin{cases} dY_t = -f(t, Y_t, Z_t)\, dt + Z_t\, dW_t, \\ Y_T = \xi \end{cases}$$

where $\xi \in L^2(\mathcal{F}_T)$

(Y, Z) is a solution if:

- Y is adapted and continuous
- Z is progressively measurable

$$\int_0^T |f(s, Y_s, Z_s)| ds + \int_0^T |Z_s|^2 ds < \infty$$

$$Y_t = \xi + \int_t^T f(s, Y_s, Z_s) ds - \int_t^T Z_s dW_s$$

Theorem (Pardoux–Peng, 1990)
Assume:

- f is globally Lipschitz in (y, z), uniformly in (ω, t);
- $\mathbb{E}\left[|\xi|^2 + \int_0^T |f(s,0,0)|^2 ds\right] < \infty.$

Then the BSDE admits a unique solution (Y, Z) with

$$\int_0^T |Z_s|^2 ds < \infty \quad \text{a.s.}$$

Example (Black–Scholes)
We consider $(S_t)_{t\geq 0}$ the following stochastic process and $C(t, S_t)$ the following conditional expectation :

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad Y_t = C(t, S_t) = \mathbb{E}^{\mathbb{Q}}\left[e^{-rT}(S_T - K)_+ \mid \mathcal{F}_t\right]$$

$$\Rightarrow \begin{cases} dY_t = f(Y_t)\, dt + Z_t\, dW_t \\ Y_T = (S_T - K)_+ \end{cases}$$

Where :

$$Z_t = \sigma S_t \partial_s C(t, S_t), \qquad f(y) = -ry$$

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Formulation of the problem

Expression of schemes

Euler scheme approximation :

Let $0 = t_0 < t_1 < \cdots < t_N = T$ be a time grid, we have the following approximation :

$$Y_{t_{k+1}} \approx Y_{t_k} - f(t_k, X_{t_k}, Y_{t_k}, Z_{t_k})\, \Delta t + Z_{t_k}\, \Delta W_k \quad (1)$$

Equivalently :

$$Y_{t_k} \approx Y_{t_{k+1}} + f(t_k, X_{t_k}, Y_{t_k}, Z_{t_k})\, \Delta t - Z_{t_k}\, \Delta W_k \quad (2)$$

$$\implies Y_{t_k} \approx \mathbb{E}\left[Y_{t_{k+1}} \mid \mathcal{F}_{t_k}\right] + f(t_k, X_{t_k}, Y_{t_k}, Z_{t_k})\, \Delta t$$

Original BSDE :

Original Deep BSDE scheme (Weinan El, Jiequn Han)

- Y_0 is fixed as a parameter y_0
- $(Z_{t_k})_{k\geq 0}$ are approximated by $(NN_{t_k}(\theta_k))_{k\geq 0}$
- We use (1) to compute :

$$Y_T = Y_0 + \sum_{k=0}^{N-1} \left(-f(t_k, X_{t_k}, Y_{t_k}, Z_{t_k})\, \Delta t + Z_{t_k}\, \Delta W_k\right)$$

- We find the optimal set of parameter (y_0, Θ) by computing the following loss :

$$\mathcal{L}(y_0, \Theta) = \mathbb{E}\left[\left|Y_{t_N}^{y_0, \Theta} - Y_T\right|^2\right]$$

- This expectation is approximated using Monte Carlo simulation over a batch of independent trajectories.

Formulation of the problem

Expression of schemes

$S_{t_k}^{(1)}, S_{t_k}^{(2)}, \dots, S_{t_k}^{(n)}$

$$Y_{t_{k+1}}^{(i)} = Y(t_k, S_{t_k}^{(i)}) - f(t_k, S_{t_k}^{(i)}, Y(S_{t_k}^{(i)}), Z(S_{t_k}^{(i)}))\Delta t + Z(t_k, S_{t_k}^{(i)})\Delta W_k^{(i)}$$

$Y_{t_{k+1}}^{(1)}, Y_{t_{k+1}}^{(2)}, \dots, Y_{t_{k+1}}^{(n)}$

DBDP$_1$ scheme (Huré, Pham, Warin)

- $Y_{t_k} \sim NN_{\theta_{Y_k}}^{Y,k}(t_k, S_{t_k}) \qquad Z_{t_k} \sim NN_{\theta_{Z_k}}^{Z,k}(t_k, S_{t_k})$
- For every step t_k we compute :

- **Target** $\widehat{Y}_{t_{k+1}}^{(i)}$ with the calibrated $(NN^{Y,k+1}, NN^{Z,k+1})$
- **Prediction** $Y_{t_{k+1}}^{(i)}$ with the new $(NN^{Y,k}, NN^{Z,k})$

- We recursively optimize the new parameters $(\theta_{Y_k}, \theta_{Z_k})$ with the local loss function :

$$\mathcal{L}_k = \mathbb{E}\left[\left|\widehat{Y}_{t_{k+1}} - Y_{t_{k+1}}\right|^2\right] \approx \frac{1}{n}\sum_{i=1}^{n}\left|\widehat{Y}_{t_{k+1}}^{(i)} - Y_{t_{k+1}}^{(i)}\right|^2$$

DBDP$_2$ scheme (Huré, Pham, Warin)

- $Y_{t_k} \sim NN_{\theta_{Y_k}}^{Y,k}(t_k, S_{t_k}) \qquad Z_{t_k} \propto \nabla_S Y(t_k, S_{t_k})$
- For every step t_k we compute :

- **Target** $\widehat{Y}_{t_{k+1}}^{(i)}$ with the calibrated $NN^{Y,k+1}$
- **Prediction** $Y_{t_{k+1}}^{(i)}$ with the new $NN^{Y,k}$

- We recursively optimize the new parameters $(\theta_{Y_k}, \theta_{Z_k})$ with the local loss function :

$$\mathcal{L}_k = \mathbb{E}\left[\left|\widehat{Y}_{t_{k+1}} - Y_{t_{k+1}}\right|^2\right] \approx \frac{1}{n}\sum_{i=1}^{n}\left|\widehat{Y}_{t_{k+1}}^{(i)} - Y_{t_{k+1}}^{(i)}\right|^2$$

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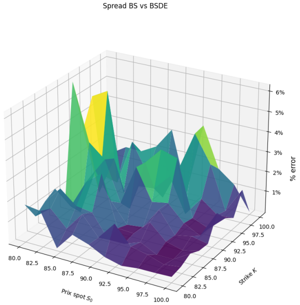
Application 1 : Study of the empirical error

European option

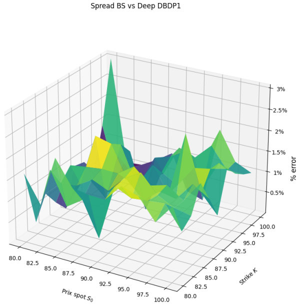
We consider $(S_t)_{t\geq 0}$ the following stochastic process and $C(t, S_t)$ the following conditional expectation :

$$dS_t = rS_t dt + \sigma S_t dW_t, \qquad Y_t = C(t, S_t) = \mathbb{E}^{\mathbb{Q}}\left[e^{-rT}(S_T - K)_+ \mid \mathcal{F}_t\right]$$

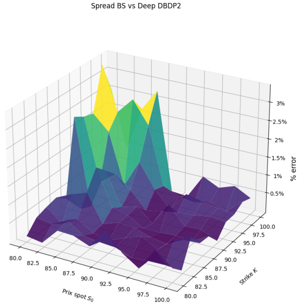
$$\Rightarrow \begin{cases} dY_t = f(Y_t)\, dt + Z_t\, dW_t \\ Y_T = (S_T - K)_+ \end{cases} \quad \text{where} \quad \begin{cases} Z_t = \sigma S_t \partial_s C(t, S_t), \\ f(y) = -ry \end{cases}$$



(a) BSDE original



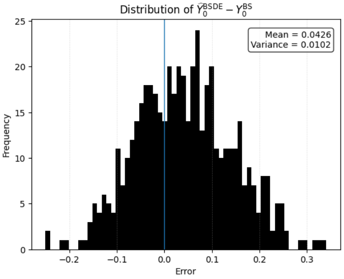
(b) Scheme $DBDP_1$



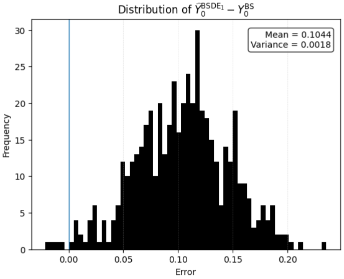
(c) Scheme $DBDP_2$

Application 1 : Study of the empirical error

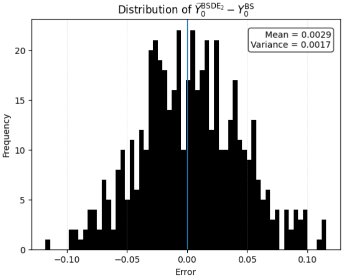
European option



(a) BSDE original



(b) Scheme $DBDP_1$



(c) Scheme $DBDP_2$

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Application 2
CVA estimation with Deep BSDE

Financial environment dynamic

Let $(W_t)_{t\geq 0} = (W_t^S, W_t^B, W_t^C)$ be a three-dimensional Brownian motion defined on a filtered probability space under the risk-neutral measure $\mathbb{Q}$.

Stock dynamic :

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t^S$$

Default intensities dynamics :

- $d\lambda_t^B = \kappa_B(\theta_B - \lambda_t^B)\,dt + \eta_B\sqrt{\lambda_t^B}\,dW_t^B$
- $d\lambda_t^C = \kappa_C(\theta_C - \lambda_t^C)\,dt + \eta_C\sqrt{\lambda_t^C}\,dW_t^C$

Instantaneous covariance matrix :

$$\Gamma_W = \begin{pmatrix} 1 & \rho_{SB} & \rho_{SC} \\ \rho_{SB} & 1 & \rho_{BC} \\ \rho_{SC} & \rho_{BC} & 1 \end{pmatrix}.$$

CVA modelling

CVA original expression :

$$\text{CVA} = \mathbb{E}^{\mathbb{Q}}\left[\int_0^T e^{-\int_0^t r_s\,ds}(1 - R_C)\,V_t^+\,\lambda_t^C\,e^{-\int_0^t(\lambda_s^B+\lambda_s^C)\,ds}\,dt\right]$$

BSDE reformulation :

$$Y_t = \mathbb{E}^{\mathbb{Q}}\left[\int_t^T e^{-\int_t^u c_s\,ds}\,g_u\,du \;\middle|\; \mathcal{F}_t\right], \qquad Y_T = 0$$

where

$$c_t = r_t + \lambda_t^B + \lambda_t^C, \qquad g_t = (1 - R_C)\,\lambda_t^C\,V_t^+.$$

$$\Longrightarrow \boxed{dY_t = \left((r + \lambda_t^B + \lambda_t^C)Y_t - (1 - R_C)\lambda_t^C V_t^+\right)dt + Z_t \cdot dW_t}$$

With : $Z_t = \left(\sigma S_t\,\partial_S Y,\; \eta_B\sqrt{\lambda_t^B}\,\partial_{\lambda^B}Y,\; \eta_C\sqrt{\lambda_t^C}\,\partial_{\lambda^C}Y\right)(t, S_t).$

Application 2 : CVA estimation with Deep BSDE

Presentation of results

Method	Mean CVA	Std CVA	Time (s)
Original scheme	0.1651	0.0095	260.3
Scheme 1	0.1693	0.0023	75.6
Scheme 2	0.1704	0.0030	89.8

Table 1. Comparison of Deep BSDE schemes.

S_0	K	T	r	R	$(\lambda_0^B, \lambda_0^C)$	σ	(κ_B, κ_C)	(θ_B, θ_C)
100	100	1	0.02	0.4	(0.02, 0.03)	(0.2, 0.2, 0.2)	(1.0, 1.5)	(0.02, 0.03)

Table 2. Table of parameters.

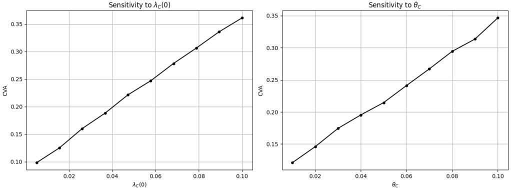


Figure: Sensitivity with respect to CIR parameters

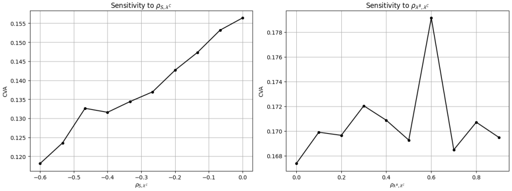


Figure: Sensitivity with respect to correlations

Extension : Deep BSDE

Extension research

Wiener Chaos decomposition (Xi Geng : 2020 2021)

Let $Y_{t_{k+1}} \in L^2(\mathcal{F}_{t_{k+1}})$. Since $\mathcal{F}_{t_{k+1}} = \mathcal{F}_{t_k} \vee \sigma(\Delta W_k)$, we have the orthogonal decomposition in L^2:

$$\boxed{L^2(\mathcal{F}_{t_{k+1}}) = L^2(\mathcal{F}_{t_k}) \oplus \bigoplus_{m \geq 1} \mathcal{H}_m^{(k)}}$$

where $\mathcal{H}_m^{(k)}$ is the chaos of order m generated by ΔW_k.

Hermite expansion

Define : $G_k := \dfrac{\Delta W_k}{\sqrt{\Delta t}} \sim \mathcal{N}(0, 1)$.

Then there exists a unique decomposition in L^2:

$$Y_{t_{k+1}} = \sum_{m=0}^{\infty} a_k^{(m)} H_m(G_k),$$

where:

- H_m are Hermite polynomials,
- $a_k^{(m)} \in L^2(\mathcal{F}_{t_k})$,
- $(H_m(G_k))_{m \geq 0}$ is an orthogonal family.

Second-order chaos approximation

Let $H_0(x) = 1, \quad H_1(x) = x, \quad H_2(x) = x^2 - 1.$

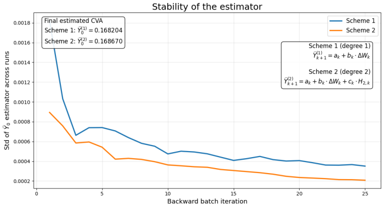
We approximate $Y_{t_{k+1}} \in L^2(\mathcal{F}_{t_{k+1}})$ by the truncated expansion

$$Y_{t_{k+1}} \approx a_k + b_k \Delta W_k + c_k((\Delta W_k)^2 - \Delta t), \quad a_k, b_k, c_k \in L^2(\mathcal{F}_{t_k}).$$

Orthogonality conditions :

$$a_k = \mathbb{E}[Y_{t_{k+1}} \mid \mathcal{F}_{t_k}], \qquad b_k = \frac{1}{\Delta t} \mathbb{E}[Y_{t_{k+1}} \Delta W_k \mid \mathcal{F}_{t_k}],$$

$$c_k = \frac{1}{2\Delta t^2} \mathbb{E}\left[Y_{t_{k+1}} ((\Delta W_k)^2 - \Delta t) \mid \mathcal{F}_{t_k}\right].$$



Pytorch

Thank you for listening!
Any questions?